DP Graded Quiz M6

Question 1

Which can be a reason to use local-volatility models instead of stochastic volatility ones?

- Stochastic volatility models require very specific (and narrow) assumptions for the volatility form.
- Local volatility models are less computationally demanding and easier to calibrate.
- Local volatility models are harder to calibrate but usually deliver better results.
- Local volatility models perform better than stochastic volatility ones.

Question 2

(**Note:** Use Python to solve this question.) Suppose you have the following data in order to calibrate your CEV model to observed market prices for different strikes (K):

- $S_0 = 64$
- $r = 0.02$
- $T = 6/12$ (6 months)
- $K = [60, 62, 63, 64]$
- Market prices for Call options (in order of strikes): $[4.64, 2.84, 2.07, 1.42]$
- Initial guess for $\sigma = 0.25$
- Initial guess for $\beta = 1.05$

Use the function error defined in the notes together with the minimize module of scipy to calibrate the CEV model parameters σ and β. What values do you obtain?

- $\sigma = 0.45; \beta = 1.27$
- $\sigma = 0.37; \beta = 1.11$

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- $\sigma = 0.40; \beta = 1.07$
- $\sigma = 0.25; \beta = 1.05$

Question 3

The Newton-Raphson method:

- consists of a root-finding algorithm based on a recursive process.
- is an alternative to model calibration.
- requires at least two free parameters in the model to match option prices.
- is generally able to find the exact price match under the Black-Scholes model.

Question 4

Select the **false** statement regarding the advantages of parametric models over non-parametric ones.

- They do not require market calibration.
- They are easier to implement.
- They involve lower computational costs.
- The implied volatility smile/surface is easier to interpolate.

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Question 5

In order to construct our local-volatility model we would, at least, need data on options:

- Market-quoted prices and strike
- Market-quoted prices
- Market-quoted prices, maturity, and strike
- Maturity and strikes

Question 6

(**Note:** Use Python to solve this question.) Suppose you have the following data in order to calibrate your CEV model to observed market prices for different strikes (K):

- $S_0 = 85$
- $r = 0.02$
- $T = 6/12$ (6 months)
- $K = [80, 82, 84, 86]$
- Market prices for Call options (in order of strikes): $[5.81, 3.92, 2.27, 1.06]$
- Initial guess for $\sigma = 0.25$
- Initial guess for $\beta = 1.05$

Use the function error defined in the notes together with the minimize module of scipy to calibrate the CEV model parameters σ and β. What values do you obtain?

- $\sigma = 0.32; \beta = 1.107$
- $\sigma = 0.27; \beta = 1.20$
- $\sigma = 0.37; \beta = 1.112$
- $\sigma = 0.34; \beta = 1.104$

Question 7

The existence of a volatility smile highlights one inherent problem of option pricing. Which one?

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- The Black-Scholes model, where volatility is constant, fails to account for volatility that varies across different strikes.
- We cannot rely on implied volatility for pricing options, as it will deliver differential values for different strike levels.
- Complex models fail to capture the volatility smile, making Black-Scholes our preferred choice.
- The need to use a model such as Black-Scholes, which perfectly captures the constant component of an underlying asset's volatility, instead of more complex models.

Question 8

Given the following CEV model SDE:

$$dS = rSdt + \sigma S^{\beta/2} dZ$$

If $\beta = 2$...

- Stock prices and volatility are unrelated.
- Stock prices and volatility relate positively.
- Stock prices and volatility relate negatively.
- The CEV model is equivalent to Black-Scholes.

Question 9

Regardless of what model we use, what is important in our model for pricing and hedging?

- Closed-form (analytical) solutions for the vanilla option price
- Fitting the real-world distribution of underlying returns
- Computational simplicity
- Consistency with market dynamics and vanilla prices

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QUESTIONS

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Question 10

(**Note:** Use Python to solve this question.) Under the CEV model closed-form equation for the call option price and using the following parameter values:

- $S_0 = 632$
- $r = 0.05$
- $T = 6/12$ (6 months)
- $K = ATM$
- $\sigma = 0.45$
- $\beta = 1.35$

What is the price of the call option?

- $19.468
- $5.782
- $0.823
- $3.863

Question 11

Which of the following models is most commonly used to extract implied volatility?

- Dupire's Equation
- Heston model
- Black-Scholes
- Constant Elasticity of Variance (CEV) model

Question 12

Given the following CEV model SDE:

$$dS = rSdt + \sigma S^{\beta/2} dZ$$

What is the return variance?

- $\sigma S^{\beta/2}$
- $\sigma^2 S^{\beta-2}$

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- $\beta - 2$
- β

Question 13

The following form for a diffusion process corresponds to which type of model?

$$dv(t) = \alpha(S, v, t)dt + \eta\beta(S, v, t)\sqrt{v(t)}dW_2$$

- Black-Scholes
- General Local volatility
- Parametric local volatility
- Stochastic volatility

Question 14

(**Note:** Use Python to solve this question.) Under the CEV model closed-form equation for the call option price and using the following parameter values:

- $S_0 = 100$
- $r = 0.05$
- $T = 3/12$ (3 months)
- $K = 105$
- $\sigma = 0.35$
- $\beta = 1.25$

What is the price of the put option?
Hint: Use put-call parity!

- $5
- $0.184
- $3.880
- $1.432

Question 15

(**Note:** Use Python to solve this question.) Assume you know the current market price of a call option is $22. The option is ATM and has time to maturity equal to 6 months. The current

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underlying asset price is $100. The risk-free rate is 3%. Using a Newton-Raphson algorithm like the one we covered, with a tolerance level of 0.0001, how many iterations does it take for the algorithm to converge?

- Three iterations
- Four iterations
- Two iterations
- More than four iterations

Question 16

Which of the following would make you decide to use a non-parametric local volatility model instead of a parametric one?

- You are worried about large computational costs.
- You do not care about the specific parameter values that define the volatility function.
- You need the numerical values for the different specific parameters that define the volatility function.
- You do not really care about the computational cost involved in the process.

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